

Demonstrating fastForce: Real-Time Reinforcement of Laser-Cut Structures

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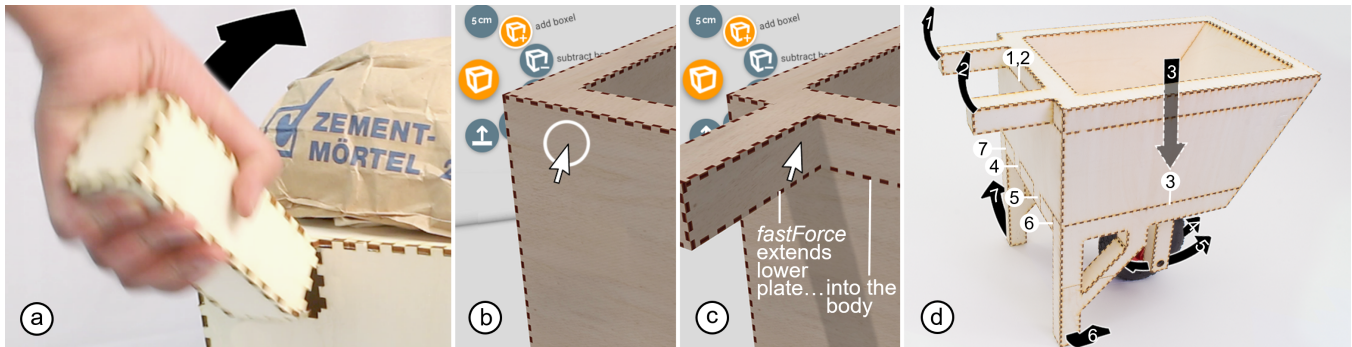


Figure 1: (a) The handle of this laser-cut wheelbarrow breaks, when the user is trying to lift a heavy load. (b) Our software tool fastForce addresses this. The moment the user clicks in the editor to create the handle, (c) not only the handle appears, but fastForce also detects the resulting point of failure and resolves it in real-time by extending the lower plate of the handle into the internal space of the wheelbarrow. (d) The resulting model has the same outer shape as the original model, yet allows lifting loads that are 45x times higher. Note that fastForce has detected additional points of failure (black arrows 2-7) and has generated corresponding reinforcement (white 2-7).

ABSTRACT

We demonstrate fastForce, a software tool that detects structural flaws in laser cut 3D models and fixes them by introducing additional plates into the model, thereby making models up to 52x stronger. By focusing on a specific type of structural issue, i.e., poorly connected sub-structures in closed box structures, fastForce achieves real-time performance. This allows fastForce to fix structural issues continuously in the background, while users

stay focused on editing their models and without ever becoming aware of any structural issues.

In our study, six of seven participants inadvertently introduced severe structural flaws into the guitar stands they designed. Similarly, we found 286 of 402 relevant models in the kyub [1] model library to contain such flaws. We integrated fastForce into a 3D editor for lasercutting (kyub) and found that even with high plate counts fastForce achieves real-time performance.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

KEYWORDS

Personal fabrication, lasercutting, structural analysis, structural reinforcement

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1 INTRODUCTION

Recently, researchers presented software systems that help users design 3D models for lasercutting, such as FlatFitFab [3], which helps users create interlocking cross sections. Kyub [2] builds on ideas from FlatFitFab and extends it towards structures that withstand 1000x larger loads. Kyub achieves this by producing “closed box structures” held together by box joints (finger joints if not perpendicular). As a result, kyub allows users to make chairs that people can sit on, even from comparably thin building materials.

Unfortunately, as the kyub authors acknowledge, in order to allow them to take such high loads, some models require “reinforcement”, i.e., they require additional internal plates to be added to the model. While the kyub authors propose users add these reinforcement plates manually, our user study finds that users are unable to do so. Similarly, we found 286 of 402 relevant user generated models in the kyub repository to contain such flaws. To illustrate this point, we encourage the reader to try to locate at least four easy ways of breaking the snowboarder sculpture shown in Figure 2a.

If users cannot figure out the need for reinforcement, reinforcement needs to be created automatically.

At first glance, such reinforcement functionality could be integrated into the user’s workflow as a separate step *after* the design workflow. However, it is generally understood that such a design-first-then-engineer workflow leads to suboptimal results. For illustration consider Figure 3a where a designer is working on two competing designs of a guitar stand. (a) The designer likes this first design for its aesthetic qualities, but (b) keeps a backup design as fallback in case the preferred design is not sturdy enough. If the environment had real-time structural analysis (e.g. as seen in [4]), the user would see warnings *during* modelling. In this example, however, there are no warnings (and if we (c) flip the preferred guitar stand around we may see why, which is that it has already been reinforced automatically by our software). The absence of a warning allows the user to dismiss the fallback design early on and (d) start elaborating on the preferred design instead.

In this paper, we address these issues by generating reinforcement (1) automatically and (2) in real-time. While we started by exploring the traditional approach of finite element analysis, we found it to be too slow for interactive use. Thus we have classified critical issues using six primitive sub-structures and developed a custom algorithm that finds and fixes points of failure in real-time by formulating them as a graph connectivity problem.

As shown in Figure 1b and c, our software tool fastForce runs in the background while users are editing their 3D model (here in kyub [2]). The moment the user starts modeling, in this case adding a handle for the wheelbarrow, fastForce simultaneously adds the necessary reinforcement, here by extending the handle’s lower plate into the interior of the wheelbarrow. (d) Since fastForce

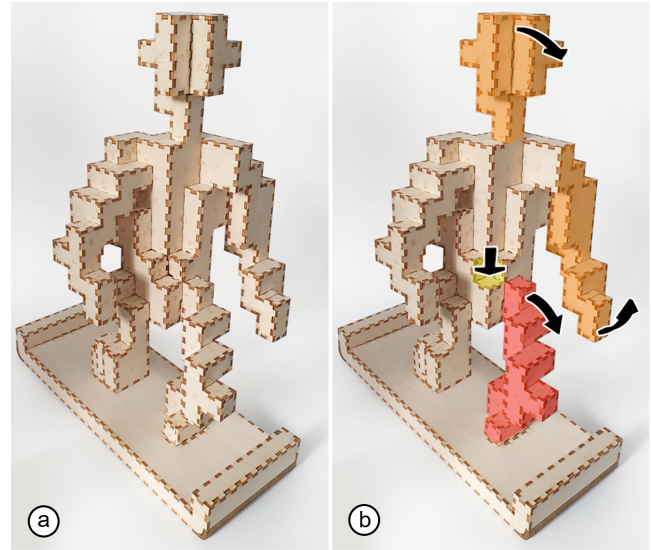


Figure 2: (a) Can you locate points of failure without looking at (b)? Here are four points of failure fastForce identifies for this model: pulling/pushing any of the red or orange parts in the direction suggested by the arrows will cause the respective part to fall inwards or rip off the rest of the model. FastForce identifies all four such points of failure, we well as another six only visible from the backside.

only modifies the wheelbarrow’s interior, the outer shape of the model is preserved while still allowing to lift a load that is 45.6 times higher. FastForce generates reinforcement for the other six points of failure similarly (black arrows 2-7). We refer to our full paper [1] for complete technical details, survey and user study.

2 INTERACTION MODEL AND INTEGRATION INTO 3D EDITOR

We integrated fastForce into a 3D editor for lasercutting (kyub [2]), allowing fastForce to be used interactively. Figure 4 illustrates this with an example of a user designing a wall mount for a bench. (a) After *stretching* the kyub’s default boxel into a bigger box, (b) the user adds an extrusion using the *add boxel* tool, but rather than showing the extrusion in isolation, (c) the reinforcement appears instantaneously with the extrusion. (d) As the user adds another extrusion, the system adds reinforcement for that as well, and combines them together. (e) When the user connects the two extrusions, (f) FastForce responds instantaneously by re-orienting the reinforcement to prevent the interlocking loop, allowing the model to be (g) laser-cut and assembled.

The key contribution we want to showcase in this walkthrough, is that there is *no* fastForce-specific user interface. Reinforcement shows up without any specific user interaction. Users can *tell* that this model has been reinforced, based on seeing the additional finger joints. These finger joints also serve as a user interface for removing the reinforcement, i.e., users may click the additional finger joints with kyub’s minus tool to delete the reinforcement in the rare case that users need control over the inside of the model (boxes,

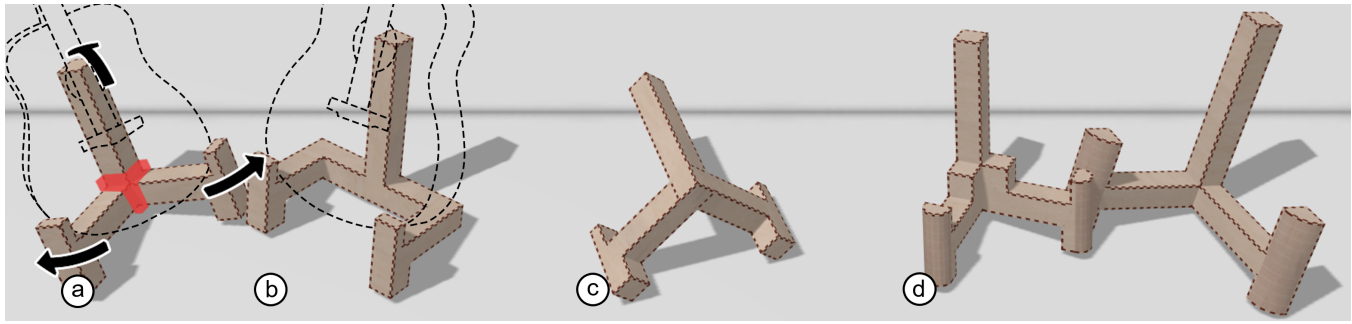


Figure 3: (a) The user’s preferred guitar stand design competes with (b) their backup design. (c) The preferred model shows no warnings (and flipping it around shows why: it has already been reinforced automatically using our system *fastForce*). (d) This allows the user to drop the fallback design and focus the design exploration on the preferred design.

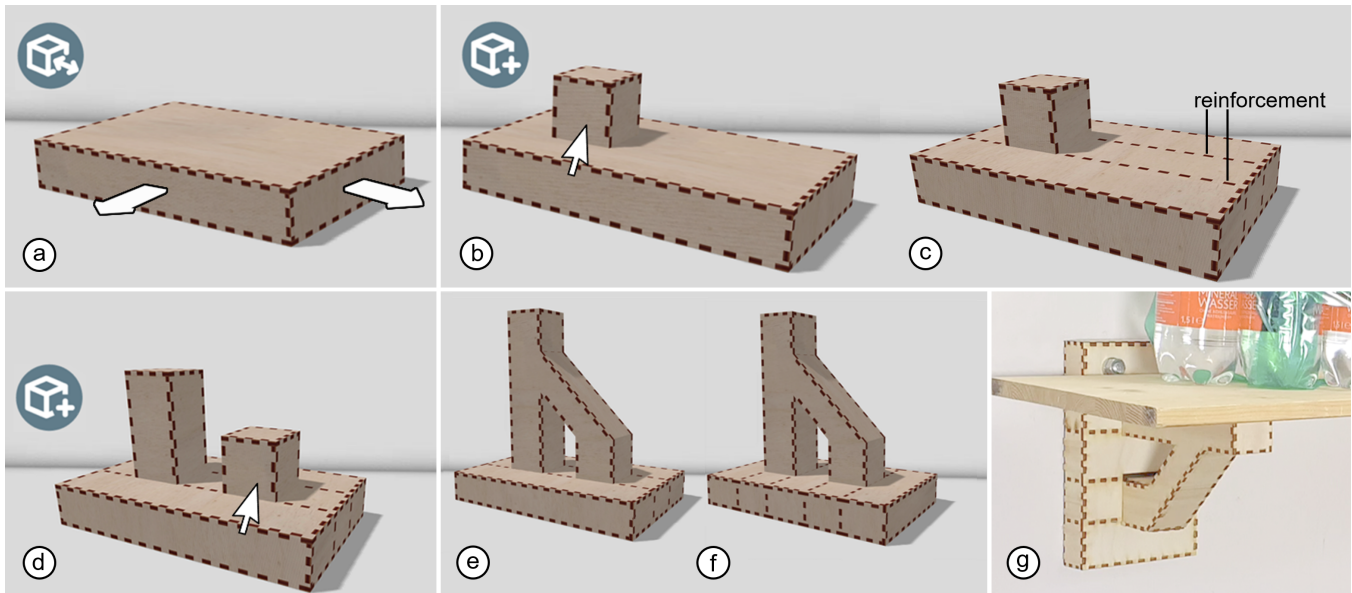


Figure 4: User session showing the creation of a wall mount using Kyub with *fastForce* integration.

containers, or molds). This turns off the automatic reinforcement so that it does not try to insert the plate again.

3 TESTING THE STRUCTURAL STRENGTH OF MODELS REINFORCED USING FASTFORCE

To evaluate the effect reinforcement created by *fastForce*, we tested the structural strength of 8 models (shown in Figure 6) after reinforcing them using *fastForce*.

We manufactured all models from standard 4mm 3-layer poplar plywood. We then used the custom rig with force sensor *forceX 2.30* made from aluminum profiles shown in Figure 5. In case of the extrusion models, we applied a bending moment from the side to the top of the extrusion. For models with a cavity, we applied a force from above the cavity to push them inwards. In both cases this results in tension forces acting at the finger joints susceptible

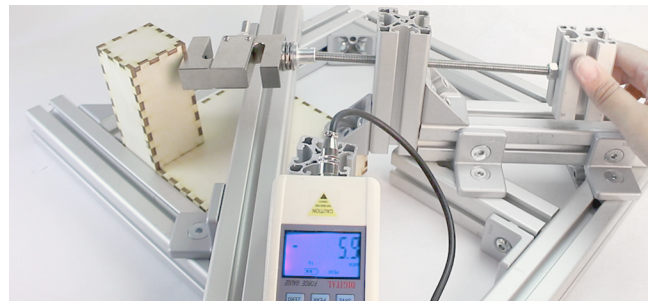


Figure 5: The test apparatus.

to tension. For the wheelbarrow and the wall mount, we applied force in the direction of their intended use case.

Results: Figure 6 shows the results. While reinforcement made the reasonably connected sub-structures E_2 and C_2 even stronger

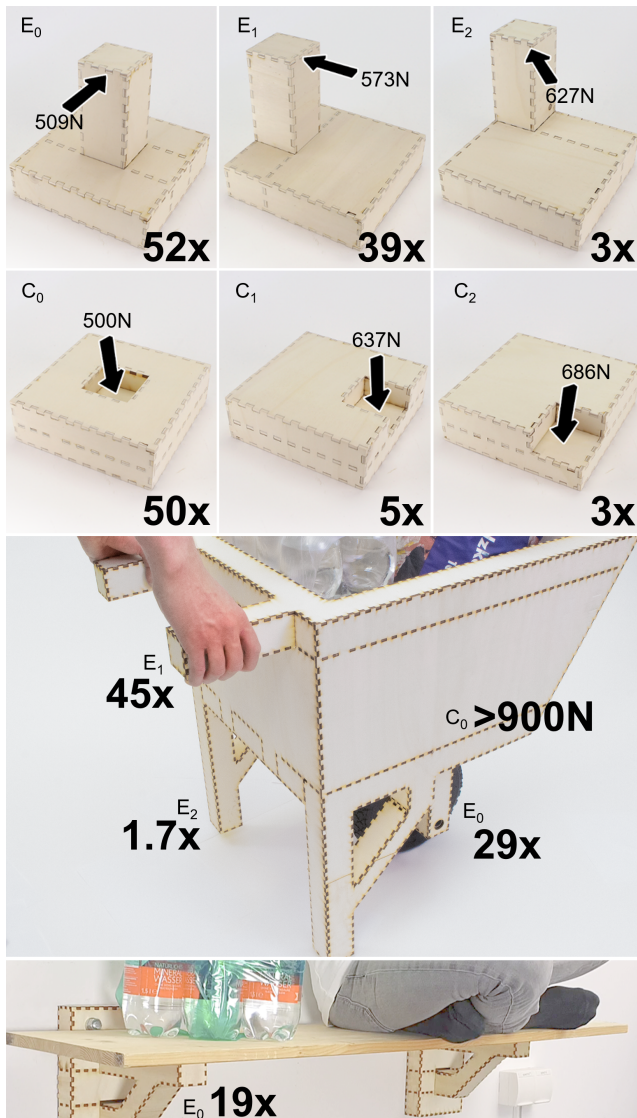


Figure 6: The reinforced test objects. Numbers indicate the factor of increase in strength after fastForce's reinforcement.

by 3x, as expected reinforcement using fastForce showed its biggest effect on the C_0 and E_0 designs, where they led to *improvements of 50x and 52x* respectively. These results confirm the fact that reinforcing laser-cut closed box structures using parallel plates considerably increases their structural strength.

4 CONCLUSION

In this demonstration, we presented fastForce, a software tool designed to identify and remove structural weaknesses in laser-cut objects based on closed box structures. We identified the fact that non-engineers are unable to identify points of failure in laser-cut closed box structures. We integrated fastForce into the interactive 3D editor kyub, where fastForce's real-time ability allows it to run in the background without interfering with users' modeling activity. Finally, we evaluated the reinforcements added by fastForce and found that reinforced closed box structures can withstand up to 52x larger forces.

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